

ECG SIGNAL ANALYSIS

OBJECTIVE: Analyse real-world ECG signal data and apply biomedical signal processing concepts to understand heart activity, signal noise, and system interactions.

THEORY: An Electrocardiogram (ECG or EKG) is a medical test that records the electrical activity of the heart over time. It is a non-invasive procedure that helps diagnose heart conditions such as arrhythmias, heart attacks, and other cardiac abnormalities. The ECG machine uses electrodes placed on the skin to detect electrical signals from the heart, which are then displayed as a waveform on a monitor or printed on paper.

The key components of an ECG waveform include the P wave (atrial depolarization), QRS complex (ventricular depolarization), and T wave (ventricular repolarization).

ECGs are commonly used in hospitals, clinics, and emergency settings. They are quick, painless, and essential for monitoring heart health. Abnormal ECG readings can indicate heart disease, electrolyte imbalances, or other medical conditions requiring further evaluation.

- 1.DATA EXPLORATION:

% 1. Data Exploration

```
%Loading ECG data and printing the first five rows
data = readtable('104samples.csv');
sig = data.x_V5_;
fs = 360;
t = (0:length(sig)-1) / fs;
disp(head(data,5));
```

We have loaded the data from the CSV file and printed the first five rows of the data present in the File.

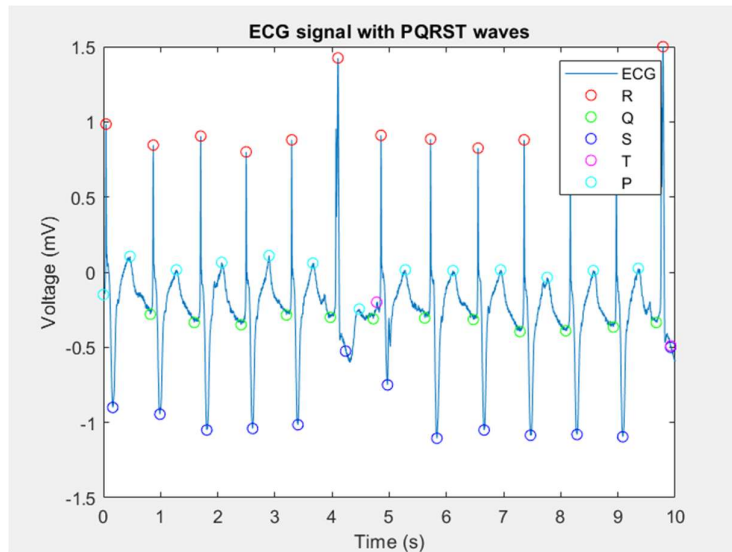
OUTPUT:

x_ElapsedTime_	x_V5_
{ '0:00.000' }	-0.15
{ '0:00.003' }	-0.15
{ '0:00.006' }	-0.15
{ '0:00.008' }	-0.15
{ '0:00.011' }	-0.15

This is the output of the first part

The key signal parameters available are “Elapsed Time” and “Amplitude of Voltages”.

• 2.ECG SIGNAL BEHAVIOUR ANALYSIS



ECG Signal with P, Q, R, S, T wave points

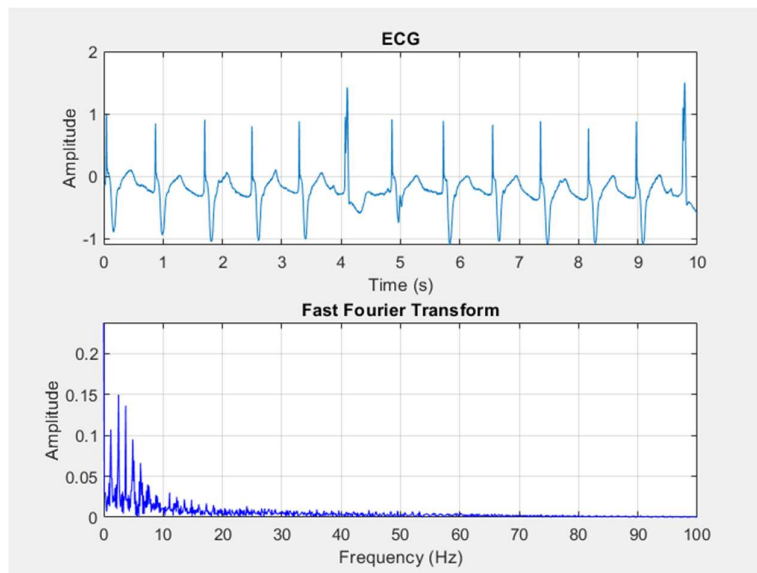
R wave: Peak with highest amplitude

Q wave: First negative deflection before R-peak

S wave: First negative deflection after R-peak

T wave: Positive deflection after S wave

P wave: Positive deflection before QRS complex



Fast Fourier Transform of ECG signal

From the FFT plot, we observe that most components of the ECG signal are of low frequency.

Calculation of average heart beat:

```
% Calculation of average heart rate(BPM) based on detected R-peaks
rr = diff(r_locs) / fs;
Heart_Rate = 60 / mean(rr);
fprintf('Heart Rate : %.2f BPM\n', Heart_Rate);
```

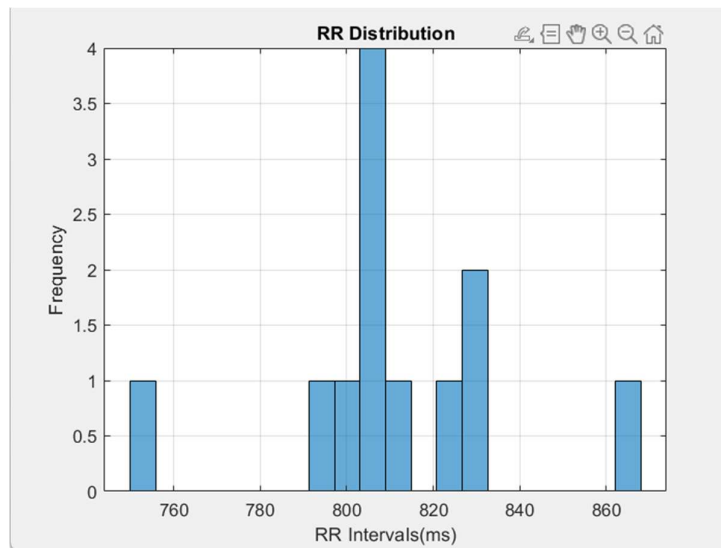
Code to calculate average heart beat using detected peaks

Here rr represents the RR intervals and diff(r_locs) gives the (index) difference between 2 successive R peaks.

Output for heart rate of the above for given dataset is:

```
Heart Rate : 73.93 BPM
```

- 3.Feature Extraction and Correlations:

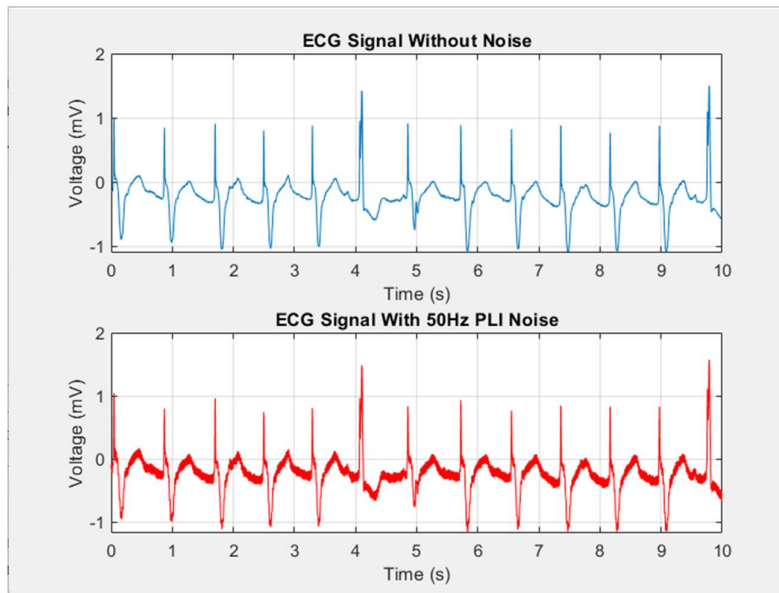


Distribution of RR intervals

Analyzing the effect of noise on the ECG waveform:

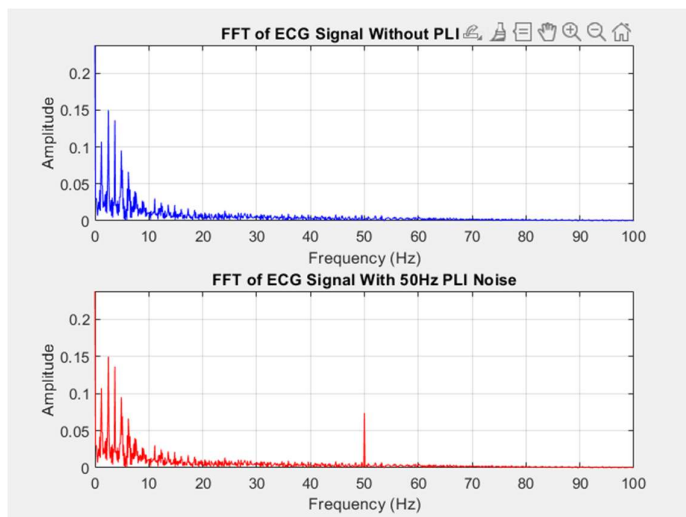
Initially there was no visible noise in the ECG waveform, so we added a Power Line Interference of 50Hz frequency.

```
f_pli = 50;  
A_pli = 0.075;  
pli_noise = A_pli * sin(2 * pi * f_pli * t)';  
noisy_sig = sig + pli_noise;
```



Plot of ECG Signal with and without Noise

It is clear from the above plot that the PLI noise has distorted the ECG waveform, so in order to get accurate readings from the ECG waveform, these noises have to be eliminated.

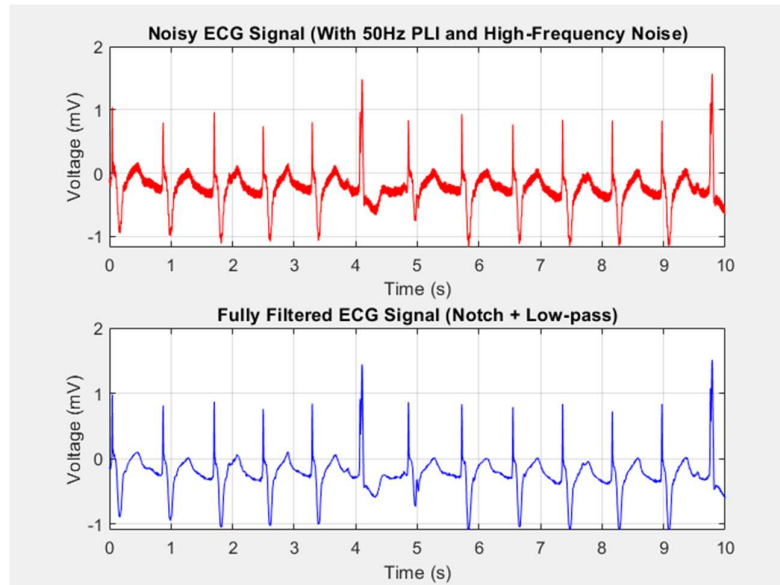


Fast Fourier Transform plot of ECG signal with and without noise

As the frequency of the PLI noise is around 50Hz, we see a sharp peak at 50 Hz in the above plot.

- **Noise Reduction and Signal Processing**

As the PLI noise is distorting the original ECG waveform, we add a filter to eliminate the effect of these noises.



Plot of ECG signal before and after filtering the PLI and other high frequency noises in from the ECG waveform.

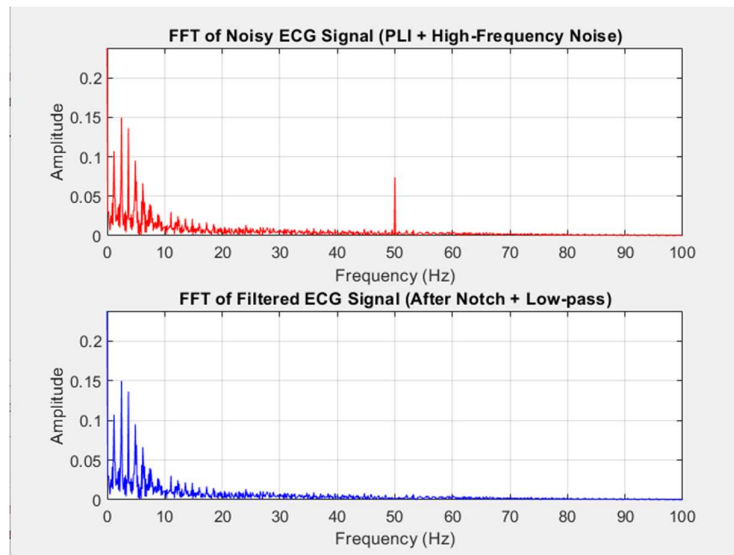
After filtering the noises, the ECG waveform is much more clear.

- **Methods to enhance ECG signal quality:**

Real time ECG monitoring requires accurate signal acquisition, noise reduction and feature extraction.

ECG signals are affected by baseline wander, PLI and motion artifacts

Filters can be used to remove these disturbances without distorting the waveform.



Plot of Fast Fourier Transform before and after filtering PLI and other high frequency noises

Calculation of mean and standard deviation of amplitudes of ECG signal before and after addition of noises

```
% Calculate mean and standard deviation after adding noise
mean_after = mean(noisy_sig);
std_after = std(noisy_sig);

fprintf('Mean before adding noise: %.4f mV\n', mean_before);
fprintf('Standard deviation before adding noise: %.4f mV\n', std_before);
fprintf('Mean after adding noise: %.4f mV\n', mean_after);
fprintf('Standard deviation after adding noise: %.4f mV\n', std_after);
```

Code to calculate mean and standard deviation of amplitude of ECG signal

```
Mean before adding noise: -0.2378 mV
Standard deviation before adding noise: 0.2764 mV
Mean after adding noise: -0.2378 mV
Standard deviation after adding noise: 0.2813 mV
```

Output of the above code for the given dataset

- Practical Implementation

If you were designing a real-time ECG monitoring system, what logic would you use to detect arrhythmias?

A) The core logic of an ECG monitoring system for arrhythmia detection would involve identifying the key features of ECG waveform like R-peak detection, R-R intervals, applying signal processing techniques to filter noise and using machine learning algorithms to classify different arrhythmia types based on the extracted features

Feature	Normal Range	Abnormalities
BPM	60 BPM – 100 BPM	Bradycardia(<50 BPM) Tachycardia(>100 BPM)
R-R interval	600ms – 1000ms	AFib(Irregular Intervals)
QRS-complex	<0.12 sec	Ventricular Tachycardia(Wide QRS complex >0.12s)
PR interval	0.12 sec – 0.2 sec	Prolongation in AV block

How can ECG signal features be used to predict cardiovascular diseases?

- A) R-R intervals, P, QRS, ST and T wave morphology, QT intervals and other features of ECG give us important heart health information. Low heart rate variability and abnormal wave patterns indicate risks like heart failures, atrial fibrillation and myocardial infarction. Training machines(models) can help us to analyze these features and predict diseases. ECG signals with other data allows us to detect cardiovascular diseases early and allow us to prevent them.

Discuss the potential of wearable ECG devices for continuous health monitoring.

- A) Wearable ECG devices have completely changed health care and have enabled us to monitor heart health in continuous and real time. Wearable ECGs allow remote patient monitoring, thereby reducing the need for frequent in person visits. This is highly helpful to older patients, patients with mobility issues or those who are living in remote areas. Wearable ECGs can be constantly used to monitor patients with chronic heart conditions, allowing health care providers to track their health overtime and adjust treatment plans as required. These devices also allow for earlier diagnosis and intervention, reducing the risk of complications like strokes.